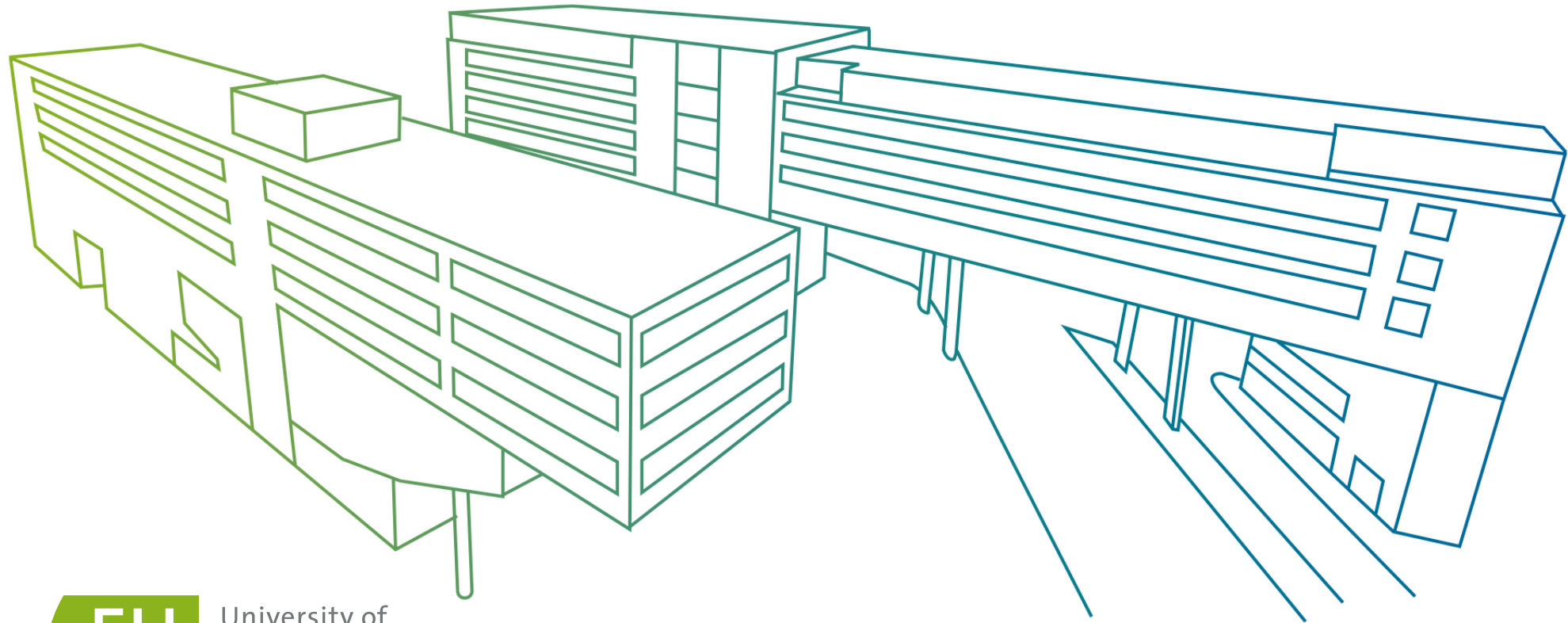




Web Technologies

CSS



Austria's Best UAS

Once again number 1 in the "Industriemagazin" ranking.



Degree Programs in English

Browse our Master's degree programs in English.



Document Check

Which documents do I need for an application?

News



07 AUGUST, 2025

First Platform Meeting



06 AUGUST, 2025

Europe Tech Hackathon



05 AUGUST, 2025

Voices from Dublin – A



01 AUGUST, 2025

Start of the First Dual

- [Study Programs](#)
- [Student Guide](#)
- [Research](#)
- [International](#)
- [About Us](#)
- [Career](#)



Degree Programs in English

Browse our Master's degree programs in English.

Newsletter Registration

Email *

☐ Yes, I would like to receive the newsletter and agree to the [Privacy Statement](#)



I'm not a robot



reCAPTCHA
[Privacy](#) • [Terms](#)

SUBMIT

Newsletter Registration

Email * ☐ Yes, I would like to receive the newsletter and agree to the [Privacy Statement](#)

Hmmmmm

Notice anything?

No style

- Our site looks like ...
- Well it doesn't look good!
- Apparently the browser has a way to "style" HTML elements, but it just looks so basic
- How can we overwrite this default style and use our own?
- Lets take the navigation bar as an example

What we write

Study Programs

Student Guide

Research

International

About Us

Career

```
<nav>
  <ul>
    <li><a href="/programs">Study Programs</a></li>
    <li><a href="/guide">Student Guide</a></li>
    <li><a href="/research">Research</a></li>
    <li><a href="/international">International</a></li>
    <li><a href="/about">About Us</a></li>
    <li><a href="/career">Career</a></li>
  </ul>
</nav>
```


What we see

- [Study Programs](#)
 - [Student Guide](#)
 - [Research](#)
 - [International](#)
 - [About Us](#)
 - [Career](#)
-
- What do we need to change?

First of all

- The links are ordered vertically not horizontally
- There is dot besides the links
 - That's because we used the `<ul>`
 - But: Lists are the correct way to structure a *list* of links in a navigation bar
- Links are blue
- Links are underlined

See what is happening

- Every browser has *Developer Tools*
 - In Chrome/Firefox it's `Ctrl + Shift + I` or `Cmd + Option + I`
 - In Safari it's the same shortcut, but you need to enable the feature in the settings first
- This is **the** tool for web development!
- You can see elements, layouts, times, events, outputs, network traffic, ... the list goes on

Let's see what our list looks like there:

Development tools

```
<nav>
  <ul>
    <li> == $0
      ::marker
      <a href="/programs">Study Programs</a>
    </li>
    <li>
      ::marker
      <a href="/guide">Student Guide</a>
    </li>
    <li>...</li>
    <li>...</li>
    <li>...</li>
    <li>...</li>
```

What can we see here? What is unexpected?

Development tools

- [Study Programs](#)
-  [Research](#)
- [International](#)
- [About Us](#)
- [Career](#)

And what can we see here?

In the browser

- The Browser adds a `::marker` to render the dot at the start of a list item
- The `<li>` has a height of 18 pixels
 - That seems to be the height of the text
- The `<li>` has a width of 580 pixels
 - That seems to be the rest of the available width

We need to change the sizing of the elements so they can fit besides each other!

Sizing

- To recap, if given no additional information, the browser applies a default size to HTML elements
- We want to add additional information to these elements so that we can change the style
- HTML is for content and structure, not for styling
- For styling we use **CSS**
 - Cascading Style Sheets

Let's add CSS to our page!

CSS

- We want to separate concerns, the HTML file should not specify the style
- We need an additional file, a CSS file, also called a stylesheet
 - Let's call it `styles.css`
- Now we need to *link* the stylesheet to the HTML document
- For that we have the `<link>` element
- Because it is not something the user needs to see, the link happens in the `<head>` of the document

CSS

```
<head>
  <title>HTML with CSS</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <!-- our nav -->
</body>
```

- the `<link>` has a `rel` and a `href` attribute
- the `href` points to the stylesheet, while the `ref` defines the relationship

Changing displays

First, we need to define what we want:

- We want to change how the `<li>` in the `<ul>` in the `<nav>` are displayed
- At the moment, the `<li>` is takes the full width with hinders other `<li>` to be next to it

There are (unfortunately) many ways to change the layout of elements in CSS, we will explore the most modern one for this requirement:

display

In the `styles.css`:

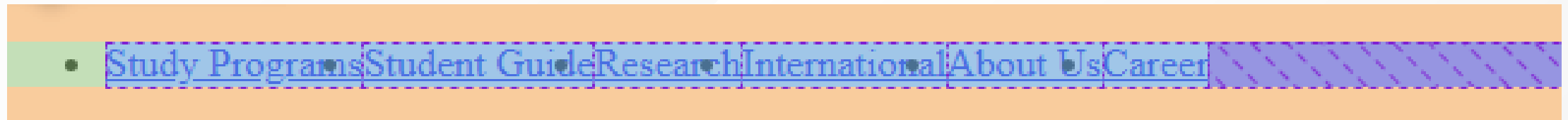
```
nav ul { /* Why "nav ul" and not just "ul"? */  
    display: flex;  
}
```

- Every HTML element has many CSS properties
- One of these properties is the `display` property
- It specifies the display behavior (the type of rendering box) of an element and the layout used for its children

Flexbox

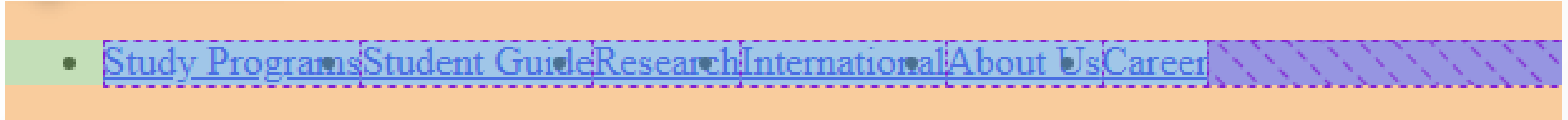
- `flex` is an efficient way to lay out, align and distribute space among items in a container
 - In our case `<ul>` is the container
 - The `<li>` are the items we want to lay out

Our CSS change results in this:



What changed?

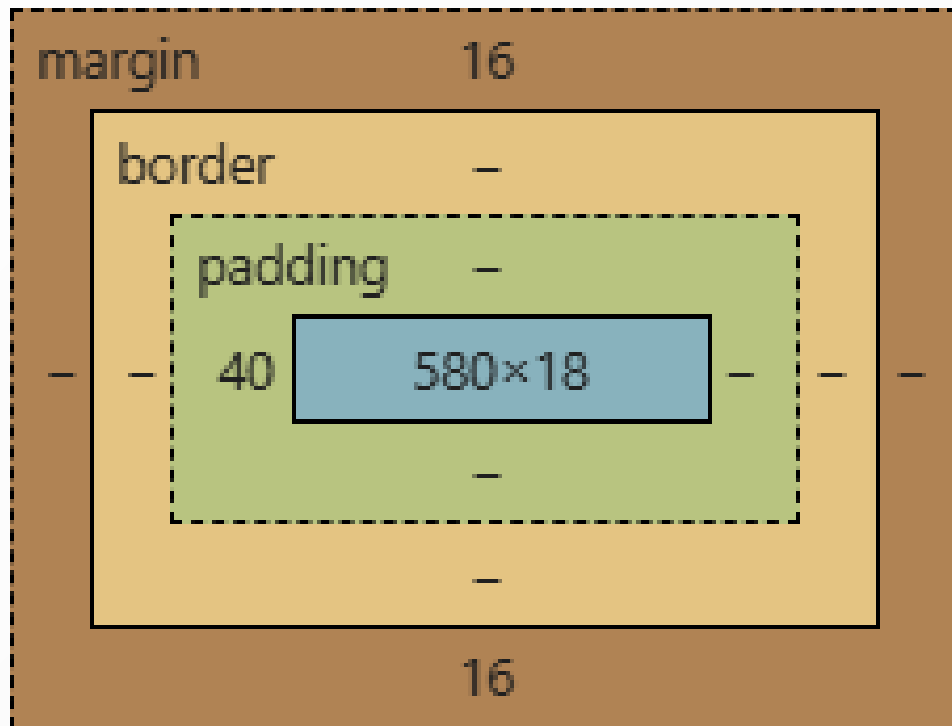
Box Model



- We can see many different colors
 - The `<ul>` is highlighted in the browser dev tools
- In orange we see the `margin` around our element
- In green we see the `padding` around our element
- In blue we see the size of our content
- In purple we see the available space
 - This is Flexbox specific

Box Model

- Every HTML element has a content size, a padding, a border and a margin.



- The browser sets default values for our elements
- We can override
 - all values e.g. `padding: 0;`
 - or individually like
`margin-top: 16px;`
`padding-left: 40px;`

Moving on

- [Study Programs](#)•[Student Guide](#)•[Research](#)•[International](#)•[About Us](#)•[Career](#)

- We achieved our first goal, the links are in one row
- But we are far from finished
- There are still those marker dots, the links are too close together and the color and font is wrong

Let's remove the dots first:

list-style

```
nav ul {  
  display: flex;  
  list-style: none;  
}
```

Every list has a `list-style` property. We don't want a list style, so we can simply set it to `none`.

[Study Programs](#)[Student Guide](#)[Research](#)[International](#)[About Us](#)[Career](#)

Let's fix the layout of the items next:

justify-content

- The items should be distributed in the line with equal space around them, except for the first and last item

```
nav ul {  
  display: flex;  
  list-style: none;  
  justify-content: space-between;  
}
```

- This only works because of flexbox!

[Study Programs](#)

[Student Guide](#)

[Research](#)

[International](#)

[About Us](#)

[Career](#)

Almost done



- There is still some lost space
- The green square of padding is blocking the space to the left

```
nav ul {  
  display: flex;  
  list-style: none;  
  justify-content: space-between;  
  padding: 0; /* removes the padding for the element */  
}
```

Layout done



The layout is done, now let's fix the colors:

- Set the color of the font to black
- Remove the underline
- Choose a sans-serif font

Navbar styling

```
nav ul {  
  display: flex;  
  list-style: none;  
  justify-content: space-between;  
  padding: 0;  
}  
  
nav ul li a {  
  color: black; /* names, RGB hexadecimal, and more are valid */  
  text-decoration: none;  
  font-family: sans-serif;  
}
```

Why did we need `nav ul li a`?

Finishing touches

Study Programs Student Guide Research International About Us Career

- It looks pretty much like the design mockup
- But there is just one little detail missing...

This is the *real* navbar after we hover our cursor over the "Student Guide" link:

Study Programs Student Guide Research International About Us Career

:hover

Another requirement:

- If our courser *hovers* over the link, we want the underline

```
nav ul li a:hover {  
    text-decoration: underline;  
}
```

- **:hover** is a CSS pseudo-class that lets you style a specific state of the selected element(s)

Done

Now our navbar looks and acts like the one in the mockup!

Mockup:

[Study Programs](#) [Student Guide](#) [Research](#) [International](#) [About Us](#) [Career](#)

Ours:

[Study Programs](#) [Student Guide](#) [Research](#) [International](#) [About Us](#) [Career](#)

The finished CSS for the navbar

```
nav ul {  
  display: flex;  
  list-style: none;  
  justify-content: space-between;  
  padding: 0;  
}  
nav ul li a {  
  color: black;  
  text-decoration: none;  
  font-family: sans-serif;  
}  
nav ul li a:hover {  
  text-decoration: underline;  
}
```


Another example

Let's quickly do the information boxes:

```
<div id="must-read">
  <div class="information">
     <!-- placeholder img -->
    <h3>Austria's Best UAS</h3>
    <p>Once again number 1 in the "Industriemagazin" ranking.</p>
  </div>
  <!-- two more information boxes --->
</div>
```

Remember the `id` and `class` attribute? Could they be useful for CSS?

How it looks without CSS

300 × 200

Austria's Best UAS

Once again number 1 in the "Industriemagazin" ranking.

300 × 200

Requirements

- Boxes not besides each other
- Text with serifs
- Text should be in grey box with white text
- boxes should have a fixed length
 - image should be the full width of the box
- There are many solutions for the layout!

Let's discuss the followings (maybe not perfect) CSS:

```
#must-read {
  display: flex;
  justify-content: space-between;
}

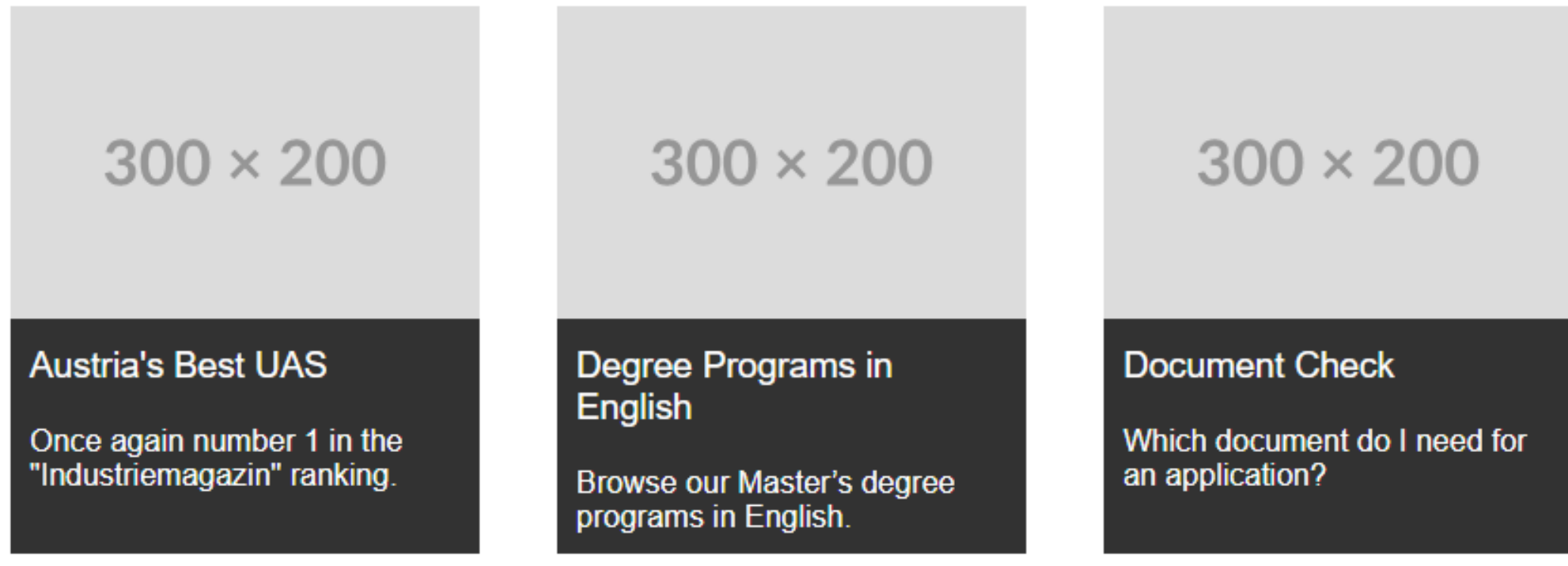
.information {
  width: 30%;
  font-family: sans-serif;
  color: white;
  background-color: #333333;
}

.information img {
  width: 100%;
}

.information h3, .information p {
  font-weight: normal;
  margin: 0;
  padding: 10px;
}
```

```
#must-read {  
    display: flex;  
    justify-content: space-between; /* we know that one */  
}  
  
.information {  
    width: 30%; /* relative value: 30% of the parent width */  
    font-family: sans-serif;  
    color: white;  
    background-color: #333333; /* RGB hex value */  
}  
  
.information img {  
    width: 100%; /* relative value: 100% of the parent (div.information) width */  
}  
  
.information h3, .information p {  
    font-weight: normal; /* font thickness */  
    margin: 0;  
    padding: 10px; /* absolute value: always 10 pixels */  
}
```

How it looks with CSS



It's not quite the same as in the mockup, but hopefully you now understand the basic idea behind CSS.

Key Takeaways

Separation of Concerns: HTML is for structure and content, CSS is for styling and presentation. Link stylesheets using `<link rel="stylesheet" href="styles.css">` in the `<head>`. Keep styling separate from HTML markup.

CSS Selectors Matter: Be specific with your selectors, e.g.: `ul` targets all lists while `nav ul` targets only lists inside nav elements. Use classes and IDs strategically. IDs are for unique elements while classes are for reusable elements.

Key Takeaways

The Box Model: Every element has content, padding, border and margin. Browsers apply (unwanted) default values that we can reset by applying a `0` value. Always remember to use browser dev tools to visualize the box model.

Modern Layout with Flexbox: `display: flex` is for flexible layouts that make responsive design (more on that later) much easier. It's perfect for navigation bars and card layouts.

Key Takeaways

CSS Values and Units: Values can be defined as absolute units (`10px`) or relative units (`30%`). Choose units based on your layout needs. It may be useful to specify an absolute width for the parent container (`<main>`) and only use relative units within this container.

Developer Tools Are Your Best Friend: Use `Ctrl/Cmd + Shift + I` to open the dev tools. You can inspect elements to see applied styles and experiment with CSS in real-time. It's essential for debugging and learning.